

p/ap dependency in oscillator

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Consider the following system with output $x(t)$:

1 Oscillator structure

$$\begin{aligned}dx &= vdt \\dv &= (-2\gamma v - \omega_0^2 x)dt + Ae^u dW_t \\du &= -\lambda udt + \sigma_u e^{\alpha u} dB_t\end{aligned}\tag{1}$$

2 your task

Give Matlab code to:

1. Simulate this system at a set of baseline parameter: $\omega_0 = 20\pi, \gamma = 5, A = 4, \lambda = 0.5, \sigma_u = 0.4, \alpha = 0.4$. Simulation duration $T = 200s$, step length $dt = 10^{-4}s$.
2. Calculate PSD of $x(t)$, fit oscillation frequency and slope of aperiodic part. Show result in figure.
3. Simulate system for $\omega_0 = 2\pi f_0$ with $f_0 = 5 : 1 : 30Hz$, show me how the oscillation frequency / oscillation strength / slope of aperiodic part change when ω_0 change.
4. Simulate system for $A = 1 : 1 : 10$, show me how the oscillation frequency / oscillation strength / slope of aperiodic part change when A change.
5. Simulate system for $\alpha = 0 : 0.1 : 0.8$, show me how the oscillation frequency / oscillation strength / slope of aperiodic part change when α change.